Polyhedra & integer programming

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Contents

All of the content is taken from [1] or [2].

I. Solving integer programs

Integer program — Definition 1

Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ and $c \in \mathbb{R}^n$. Consider index sets $I, C \subseteq [n]$ such that $I \cup C = [n]$ and $I \cap C = \emptyset$. We define the *Mixed Integer Linear Program* (MILP) as

$$\max_x \quad c^\top x \quad \text{subject to} \quad Ax \leq b, x_I \in \mathbb{Z}^I, x_C \in \mathbb{R}^C \tag{1}$$

I.1. Branch & cut

Continuous relaxation — Definition 2

The continous relaxation of the MILP from Equation 1 is the *Linear Program* (LP)

$$\max_x \quad c^\top x \quad \text{subject to} \quad Ax \leq b, x \in \mathbb{R}^n \tag{2}$$

Branch & cut — Algorithm 3

The standard algorithm for solving MILPs is branch & cut, which builds a search tree iteratively starting from the original problem:

1. Pick a node in the search tree and solve its continuous relaxation.
2. Prune the node if
   - The continuous solution is integer (update best-known integer solution)
   - The continuous solution is worst than the best-known integer solution
3. Otherwise, strengthen the formulation through cuts, excluding the continuous solution but keeping all integer solutions.
4. Branch to divide the feasible region into a partition and add corresponding nodes to the search tree.

The goal of cuts is bringing the continous LP closer to the original IP in terms of bound quality. This is key for practical efficiency of the method.

I.2. Structure of the feasible set

Convex hull doesn't matter for LPs — Remark 4

Given a set S, maximizing a linear objective on S or $\text{conv}(S)$ yields the same value.

Proof:

Obviously, $\max_{x \in S} c^\top x \leq \max_{x \in \text{conv}(S)} c^\top x$. Conversely, if $x \in \text{conv}(S)$ reaches the maximum (same reasoning applies if it is ε-close), there exist weights λ_i and vectors $x_i \in S$ such that $x = \sum_i \lambda_i x_i$. Thus, $c^\top x = \sum_i \lambda_i (c^\top x_i)$ and at least one of the x_i satisfies $c^\top x_i \geq c^\top x$.

Perfect formulation — Definition 5

A perfect formulation of a set S is a set of linear inequalities such that $\mathrm{conv}(S) = \{x : Ax \leq b\}$.

Perfect formulation for polytopes — Proposition 6

Let $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a polyhedron and $S = P \cap \mathbb{Z}^n$. If P is bounded, then $\mathrm{conv}(S)$ admits a perfect formulation.

Proof:

If P is bounded, S is finite, so it $\mathrm{conv}(S)$ is the convex hull of a finite number of points with integer coordinates. By Minkowski-Weyl, it is a bounded polyhedron.

Meyer's theorem — Theorem 7

Let $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a polyhedron and $S = P \cap \mathbb{Z}^n$. If A and b have rational coefficients, then $\mathrm{conv}(S)$ admits a perfect formulation with rational coefficients.

Key question of today: what can we say about that perfect formulation? Can we compute it exactly? Approximate it iteratively?

II. Polyhedral theory

II.1. Basics

Polyhedra — Definition 8

A *polyhedron* P is an intersection of finitely many half-spaces:

$$P = \{x \in \mathbb{R}^n : Ax \leq b\} \tag{3}$$

Cones — Definition 9

- A *cone* C is a set that is stable by positive scaling
- A *finitely generated cone* is all the conic combinations of a finite set of rays.
- A *polyhedral cone* is a cone that is also a polyhedron (defined by $b = 0$)

Polytopes — Definition 10

A *polytope* Q is the convex hull of a finite set of vectors.

Minkowski-Weyl for cones — Theorem 11

A cone C is polyhedral if and only if it is finitely generated.

Proof:

Supppose C is finitely generated. By Fourier elimination we can obtain a polyhedral representation (with exponential size)

Now suppose C is polyhedral:
1. By Farkas' lemma C^* (the dual cone) is finitely generated.
2. By the first point, C^* is polyhedral.
3. Because $\mathbb{R}^n_+$ is self-dual, C is finitely generated.

Minkowski-Weyl for polytopes — Theorem 12

A set P is a polyhedron if and only if $P = Q + C$ where Q is a polytope and C is a finitely generated cone.

Proof:

Suppose $P = \{Ax \leq b\}$. P can be represented in higher dimension as the intersection of a cone with a hyperplan: $P = \{(x, 1) : x \in C_P\}$ where

$$C_P = \{(x, y) \in \mathbb{R}^{n+1} : Ax - by \leq 0, y \geq 0\} \tag{4}$$

We know that C_P is finitely generated: $C_P = \text{cone}\{(r_1, s_1), ..., (r_k, s_k)\}$. Any $x \in P$ can be written as $x = \sum_i \mu_i r_i$ with $\mu_i \geq 0$ and $\sum_i \mu_i s_i = 1$. Thus we have

$$x = \sum_{i:s_i>0} \underbrace{\mu_i s_i}_{\lambda_i} \left(\frac{r_i}{s_i}\right) + \sum_{i:s_i=0} \mu_i r_i \tag{5}$$

Defining $v_i = \frac{\mu_i}{s_i}$, we get $P = \text{conv}\{v_k : s_k > 0\} + \text{cone}\{r_\ell : s_\ell = 0\}$.

Now suppose $P = \text{conv}\{v_k\} + \text{cone}\{r_\ell\}$ and define $C_P = \text{cone}\left(\{(v_k, 1)\} \cup \{(r_\ell, 0)\}\right)$. That cone is finitely generated, so it is polyhedral: there exist A and b such that $C_P = \{(x, y) \in \mathbb{R}^n : Ax - by \leq 0\}$. Therefore, $P = \{x \in \mathbb{R}^n : Ax - b = 0\}$.

Flow polyhedron — Exercise 13

The flow polyhedron is the sum of the polytope of paths and the cone of circulations.

Recession cone, lineality space — Definition 14

Given a non-empty polyhedron P, we define
- its *recession cone* $\text{rec}(P) = \{r \in \mathbb{R}^n : x + \lambda r \in P \text{ for all } x \in P, \lambda \geq 0\}$ (set of rays of the polyhedron)
- its *lineality space* $\text{lin}(P) = \{r \in \mathbb{R}^n : x + \lambda r \in P \text{ for all } x \in P, \lambda \in \mathbb{R}\}$.

When $\text{lin}(P) = \{0\}$, we say that the polyhedron is *pointed* (doesn't contain any line).

Links with Minkowski-Weyl Proposition 15

Let $P = \text{conv}\{v_k\} + \text{cone}\{r_\ell\} = \{x : Ax \leq b\}$ be a non-empty polyhedron. Then
- $\text{rec}(P) = \{r : Ar \leq 0\} = \text{cone}\{r_\ell\}$
- $\text{lin}(P) = \{r : Ar = 0\}$.

Proof:

- $\text{rec}(P) = \{r : Ar \leq 0\}$ is obvious: if there is i such that $a_i^\top r > 0$, we cannot scale r forever while staying $\leq b_i$ and vice-versa.
- If $r \in \text{cone}\{v_k\}$ then $\lambda r \in \text{cone}\{v_k\}$ and since $P = \text{conv}\{v_k\} + \text{cone}\{r_\ell\}$ we have $x + \lambda r \in P$ for all $x \in P$ and $\lambda \geq 0$. If $r \notin \text{cone}\{r_\ell\}$, then $\text{dist}(x, \text{cone}\{r_\ell\}) > 0$. It is easy to prove that $\text{dist}(\lambda r, \text{cone}\{r_\ell\}) = \lambda \text{ dist}(r, \text{cone}\{r_\ell\})$. For large enough λ, we will have $\text{dist}(\lambda r, \text{cone}\{r_\ell\}) > \text{diam}(\text{conv}\{v_k\})$, which prevents $x + \lambda r$ from belonging to P.
- Finally, $\text{lin}(P) = \text{rec}(P) \cap -\text{rec}(P) = \{x : Ax \leq 0\} \cap \{x : Ax \geq 0\}$.

II.2. Polyhedron dimension

Affine independence Definition 16

- Points $x_0, ..., x_q$ are *affinely independent* if the only solution to $\sum_{j=0}^{q} \lambda_j x_j = 0$ and $\sum_{j=0}^{q} \lambda_j = 0$ is $\lambda_0 = ... = \lambda_q = 0$.
- The *dimension* of a set S is the maxiumum number of affinely independent points inside S.
- The *affine hull* $\text{aff}(S)$ of a set S is the inclusion-wise minimal affine subspace containing it.

Implicit equality Definition 17

We say that $a_i x \leq b$ is an implicit inequality in the system $Ax \leq b$ if $a_i x = b_i$ is satisfied for every solution of $Ax \leq b$. We denote by $A^= x \leq b^=$ the subsystem of all implicit equalities (indexed by $I^=$) and by $A^< x \leq b^<$ the rest (indexed by $I^<$). So

$$P = \{x : A^= x = b^=, A^< x \leq b^<\} = \{x : A^= x \leq b^=, A^< x \leq b^<\} \quad (6)$$

Existence of strictly feasible point Remark 18

If $P \neq \emptyset$, then P contains a point $\overline{x}$ such that $A^< \overline{x} < b^<$.

Proof:

There is a strictly satisfying point for each constraint that is not an implicit equality. Take the average.

Affine hull — Theorem 19

Let $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a non-empty polyhedron.

$$\mathrm{aff}(P) = \{x \in \mathbb{R}^n : A^{=}x = b^{=}\} = \{x \in \mathbb{R}^n : A^{=}x \leq b^{=}\} \tag{7}$$

and $\dim(P) = n - \mathrm{rank}(A^{=})$.

Proof:

By definition, $P \subseteq \{x : A^{=}x = b^{=}\}$ and the latter is an affine space, so $\mathrm{aff}(P)$ satisfies the same inclusion. Also, $\{x : A^{=}x = b^{=}\} \subseteq \{x : A^{=}x \leq b^{=}\}$.

Conversely, let $\hat{x} \in \{x : A^{=}x \leq b^{=}\}$. We know there is a strictly satisfying point $\tilde{x} \in P$ such that $A^{<}\tilde{x} < b^{<}$ and $A^{=}\tilde{x} = b^{=}$. There exists $\varepsilon > 0$ small enough such that $z = \tilde{x} + \varepsilon(\hat{x} - \tilde{x})$ satisfies $A^{<}z \leq b^{<}$ and still $A^{=}z \leq b^{=}$, so $z \in P$. Thus, the whole line linking $\tilde{x}$ and z is contained in $\mathrm{aff}(P)$, hence $\hat{x} \in \mathrm{aff}(P)$.

And finally $\dim(\mathrm{aff}(P)) = n - \mathrm{rank}(A^{=})$.

Full-dimensional — Definition 20

A polyhedron $P \subseteq \mathbb{R}^n$ is full-dimensional if $\dim(P) = n$.

Identifying the dimension of a set — Remark 21

We give two methods to prove that $\dim(S) = k$:

1. Find a system $Ax = b$ such that $S \subseteq \{x : Ax = b\}$ and $\mathrm{rank}(A) = n - k$, and find $k + 1$ affinely independent points in S.
2. Same first step but then prove that every other equality satisfied by all $x \in S$ can be obtained from combinations of $Ax = b$ (this means that every affine space containing S also contains $\{x : Ax = b\}$)

Knapsack polytope — Exercise 22

The knapsack polytope is $P = \mathrm{conv}\{x \in \{0,1\}^n : a^\top x \leq b\}$ where $a \in \mathbb{R}^n_+$ and $b > 0$. Show that $\dim(P) = n - |J|$ where $J = \{j : a_j > b\}$.

Since $P \subseteq \{x \in \mathbb{R}^n : x_j = 0 \text{ for all } j \in J\}$, we have $\dim(P) \leq n - |J|$. On the other hand, if $j \notin J$, then $x = e_j \in P$. Since the origin is in P too, we have $n - |J| + 1$ affinely independent points in there.

Hamiltonian path polytope — Exercise 23

The Hamiltonian path polytope P of a graph $G = (V, E)$ is the convex hull of the incidence vectors of the Hamiltonian paths of G (those that go exactly once through each node). Show that if G is the complete graph on n nodes, $\dim(P) = \binom{n}{2} - 1$.

All Hamiltonian path incidence vectors x satisfy $\sum_e x_e = n-1$. Thus $\dim(P) \le \binom{n}{2} - 1$. Let $\alpha^\top x = \beta$ be any other equality satisfied by all $x \in P$. It suffices to prove that $\alpha_e = \alpha_{e'}$. Let T be a Hamiltonian tour containing both e and e'. Then $T \setminus \{e\}$ and $T \setminus \{e'\}$ are both Hamiltonian paths, and their incidence vectors satisfy $\alpha^\top x = \alpha^\top x' = \beta$, so $0 = \alpha_e - \alpha_{e'}$.

II.3. Faces

Valid inequality — Definition 24

An inequality $c^\top x \le \delta$ is *valid* for the set P if it is satisfied by every point in P.

Valid inequalities are combinations — Theorem 25

An inequality $c^\top x \le \delta$ is valid for $P = \{x \in \mathbb{R}^n : Ax \le b\}$ if and only if there exists $u \in \mathbb{R}^m_+$ such that $c = \sum_i u_i a_i$ and $\sum_i u_i b_i \le \delta$.

Proof:

If $c^\top x \le \delta$ is a nonnegative combination of $A^\top x \le b$, the inequality is obviously valid.

If $c^\top x \le \delta$ is valid, then $\max_{x \in P} c^\top x$ admits a finite optimum with value $\delta' \le \delta$. Let u be a dual optimal solution: it is such that $A^\top u = c$ and $b^\top u = \delta' \le d$.

Face — Definition 26

- A *face* of a polyhedron P is a set of the form $P = P \cap \{x : c^\top x = \delta\}$ where $c^\top x \le \delta$ is a valid inequality for P.
- If a valid inequality defines a face, the hyperplane $\{c^\top x = \delta\}$ is called a *supporting hyperplane*.
- A face F is *proper* if $F \ne \emptyset$ and $F \ne P$.
- A *facet* is an inclusionwise-maximal, proper face of P.

Draw simple exercises.

Characterization of faces — Theorem 27

Let $P = \{x : Ax \le b\}$ be a non-empty polyhedron with constraint set M. For any $I \subseteq M$ with $J = M \setminus I$,

$$F_I = \{x : A^I x = b^I, A^J x \le b^J\} \tag{8}$$

is a face of P. Conversely, any non-empty face of P is equal to F_I for some I.

Proof:

Let $I \subseteq M$. We pick $c = \sum_{i \in I} a_i$ and $\delta = \sum_{i \in I} b_i$ to define the face $F = P \cap \{x : c^\top x = \delta\}$. Obviously $F_I \subseteq P$ and any $x \in F_I$ satisfies $c^\top x = \sum_{i \in I} a_i^\top x = \sum_i b_i = \delta$, so $F_I \subseteq F$. Let $x \in F$: we know that $a_i^\top x \le b_i$ for all $i \in I$. If there was an $i \in I$ such that $a_i^\top x <$

b_i, we would have $c^\top x < \delta$, thus x would not be in F. Therefore, $x \in F_I$. We conclude that $F_I = I$, and so F is a face.

Conversely, let F be a non-empty face of P. Since the inequality is valid, xF is the set of optimal primal solutions to $\max_{x \in P} c^\top x$. Let u be an optimal dual solution to that linear program. $\min_{u \geq 0, A^\top u = c} b^\top u$ The complementary slackness condition $u_i(a_i^\top x - b_i) = 0$ holds for all $i \in M$ if and only if $x \in F$. Thus, with $I = \{i \in M : u_i > 0\}$, we have $x \in F$ if and only if $x \in P$ and $a_i^\top x = b_i$ for all $i \in I$.

Facts about faces — Proposition 28

Let P be a polyhedron.
- The set of faces is finite.
- The set of faces is stable by intersection.
- Two faces are distinct if and only if they have different affine hulls.
- If $F' \subset F$ then $\dim(F') < \dim(F)$.
- $\mathrm{lin}(F) = \mathrm{lin}(P)$

Reundant constraint — Definition 29

Given a polyhedron $P = \{x : Ax \leq b\}$, we say that a constraint i is *redundant* if removing it doesn't change the polyhedron.

When all constraints are redundant — Exercise 30

Each can be present twice.

Irredundant inequality constraints — Proposition 31

Let P be a non-empty polyhedron and $j \in I^<$ such that $a_j^\top x \leq b_j$ is irredundant. We define the face $F = \left\{x \in P : a_j^\top x = b_j\right\}$. Then
- F contains an $\hat{x}$ such that $a_i^\top x < b_i$ for all $i \in I^< \setminus \{j\}$.
- $\mathrm{aff}(F) = \{x : a_i^\top x = b_i \text{ for all } i \in I^= \cup \{j\}\}$ and $\dim(F) = \dim(P) - 1$

Proof:

There exists $\overline{x} \in P$ that is strictly feasible for all $i \in I^<$. Since j is not redundant, removing that constraint enlarges the polyhedron. There exists $\tilde{x}$ satisfying all the constraints of P except j, so that $a_j^\top \tilde{x} > b_j$. The segment $[\overline{x}, \tilde{x}]$ goes through F at, say, $\hat{x}$. The intersection satisfies $a_i^\top \hat{x} = b_i$ for all $i \in I^=$ and $a_i^\top \hat{x} < b_i$ for all $i \in I^< \setminus \{j\}$.

The system

$$a_i^\top x = b_i, i \in I^= \cup \{j\} \quad \text{and} \quad a_i^\top x \leq b_i, i \in I^< \setminus \{j\} \tag{9}$$

has a set of implicit equalities given by $I^= \cup \{j\}$. Thus its affine hull is $\mathrm{aff}(F) = \{x : A^= x = b^=\} \cap \left\{x : a_j^\top x = b_j\right\}$.

We know that F is strictly contained in P since $j \in I^<$, so $\dim(F) < \dim(P)$. We also know that $\dim(F) = n - \operatorname{rank}\big((A^=, a_j)\big) \geq n - \operatorname{rank}(A^=) - 1 = \dim(P) - 1$.

Minimal representation — Definition 32

The system

$$a_i^\top x = b_i, i \in I^= \quad \text{and} \quad a_i^\top x \leq b_i, i \in I^< \tag{10}$$

defining P is a *minimal representation* when all the constraints are irredundant.

Redundant inequalities — Exercise 33

Let $a_1^\top x \leq b_1$ and $a_2^\top x \leq b_2$ be redundant inequalities for $P = \{x : Ax \leq b\}$. We construct P' by removing $a_1^\top x \leq b_1$ from P. Is $a_2^\top x \leq b_2$ always redundant for P'?

Characterization of facets — Theorem 34

Let $P = \{x : Ax \leq b\}$ be a non-empty polyhedron.

- For each facet F of P, there is an irredundant constraint $j \in I^<$ such that $F = P \cap \left\{x : a_j^\top x = b_j\right\}$.
- For any minimal representation, P has $|I^<$ facets and each of them is associated with an irredundant inequality constraint.
- A face F of P is a facet if and only if it is non-empty and $\dim(F) = \dim(P) - 1$.

Proof:

- Let F be a facet. There exists a set of constraints $I \subseteq I^<$ such that $F = \big\{x \in P : a_i^\top x = b_i \text{ for all } i \in I\big\}$ Since $F \neq P$, $I \neq \emptyset$. Let $j \in I$ and $F' = \left\{x \in P : a_j^\top x = b_j\right\}$. F' is also a face and $F \subseteq F' \subset P$ since $j \in I^<$. By maximality, $F = F'$, so $I = \{j\}$ and F is defined by $a_j^\top x \leq b_j$.
- Each irredundant inequality constraint defines a facet. Furthermore, every facet is defined by an irredundant inequality constraint as we just saw. The last ingredient is proving that these facets are distinct, which comes from the existence of a point in each facet that strictly satisfies all other inequality constraints.

Stable set polytope — Exercise 35

The stable set polytope P is the convex hull of the characteristic vectors x of all the stable sets in a graph G.

- Find the dimension of P
- Given a maximal clique K, show that the inequality $\sum_{v \in K} x_v \leq 1$ is valid and facet-defining.

P is full-dimensional since it contains 0 and every basis vector e_i. We want to find $|V|$ affinely-independent vectors in the face F_K. We start by taking e_k for $k \in K$. For each vertex $v \in V \setminus K$, we need to pair it with $k \in K$ such that $x = e_v + e_k \in P$. This means that (v, k) should not be an edge of the graph. Luckily, if there was a $v \in V \setminus K$ connected to everyone in the clique, K would not be maximal.

Uniqueness of the minimal representation — Theorem 36

The minimal representation of a full-dimensional polyhedron is unique up to permutation and multiplication of the inequalities.

Importance of facets — Remark 37

Facets are the tightest cuts that one can find, which means they are very useful in branch & cut solvers.

II.4. Vertices & edges

Minimal face — Definition 38

A minimal face of a polyhedron is a non-empty face that contains no proper face.

Faces of affine spaces — Exercise 39

Prove that an affine space P has no proper face: given a valid inequality $c^\top x \leq \delta$, either all points in P satisfy $c^\top x = \delta$, or all points in P satisfy $c^\top x < \delta$.

Characterization of minimal faces — Theorem 40

Let $P = \{x : Ax \leq b\}$ be a non-empty polyhedron.

- A non-empty face F of P is minimal if and only if $F = \{x : A'x = b'\}$ for some subsystem $A'x \leq b'$ from $Ax \leq b$ such that $\operatorname{rank}(A') = \operatorname{rank}(A)$.
- The dimensions of non-empty faces of P range from $\dim(\operatorname{lin}(P))$ to $\dim(P)$.

Proof:

Let F be a non-empty face of P. If $F = \{x : A'x = b'\}$ then F is an affine space and has no proper face, so F is a minimal face of P. If F is a minimal face of P, let $A'x = b'$ be the subsystem of $Ax = b$ that is saturated at every point of F, and $A''x \leq b''$ the rest. Let $C^=, C^<, d^=, d^<$ be a minimal representation of F extracted from $A'x = b', A''x \leq b''$. Since F is minimal, it has no facet, so $C^< = \emptyset$ and $F = \{x : C^= x = d^=\} = \{x : A'x = b'\}$. F is an affine subspace so $F = \{v\} + \operatorname{lin}(F)$ and $\dim(F) = \dim(\operatorname{lin}(P))$, so $\operatorname{rank}(A') = n - \dim(F) = n - \dim(\operatorname{lin}(P)) = \operatorname{rank}(A)$. Finally, a face of dimension $> \dim(\operatorname{lin}(P))$ contains a facet.

Minimal faces of a polyhedral cone — Exercise 41

How many does it have?

Vertices, edges — Definition 42

- A face of dimension 0 of P is called a *vertex*.
- A face of dimension 1 of P is called an *edge* (or an *extreme ray* for a polyhedral cone).
- The *skeleton* of a polytope is the graph defined by vertices and edges.

Characterization of vertices — Theorem 43

Let $P = \{x : Ax \leq b\}$ be a pointed polyhedron and $\overline{x} \in P$. The following are equivalent:
1. $\overline{x}$ is a vertex.
2. $\overline{x}$ saturates n linearly independent inequalities of $Ax \leq b$.
3. $\overline{x}$ is not a proper convex combination of distinct points of P.

Proof:

P is pointed so $\text{lin}(P) = \{0\}$ and $\text{rank}(A) = n$.

- By the previous theorem, any vertex $\overline{x}$ is a minimal face which satisfies n linearly independent equalities from $Ax \leq b$ and vice-versa.
- Let $\overline{x}$ satisfy n linearly independent equalities, and let y, z be such that $x = \lambda y + (1 - \lambda)z$ with $\lambda \in (0, 1)$. Since $\{\overline{x}\}$ is a face, there exists a valid inequality $c^\top x \leq \delta$ saturated only at $\overline{x}$ inside P. We see that y and z are in P saturate it too, so $y = z = x$.
- Let $\overline{x}$ be such that the maximum set of independent equalities satisfied is $A'x = b$ with $\text{rank}(A') < n$, we denote by $A''x < b''$ the rest. There exists a vector $v \in \ker(A') \setminus \{0\}$ such that $A'v = 0$. Taking $y = x - \varepsilon v$ and $z = x + \varepsilon v$ with $\varepsilon > 0$ small enough gives us two points that are still in P and such that $x = \frac{y+z}{2}$, so $\overline{x}$ is not an extreme point.

Vertices of underconstrained polyhedra — Exercise 44

Does the polyhedron $P = \{x : Ax \leq b\}$ have a vertex when $m \leq n$?

Carathéodory — Theorem 45

- Let v be a conic combination of some set X: it is a conic combination of at most $\dim(\text{cone}(X))$ linearly independent vectors of X.
- Let v be a convex combination of some set X: it is a convex combination of at most $\dim(\text{conv}(X)) + 1$ affinely independent vectors of X.

Proof:

- We first prove the conic case. Let $x_1, ..., x_k$ and $\lambda_1, ..., \lambda_k$ be such that $v = \sum_i \lambda_i x_i$. The polyhedron $P = \left\{\lambda \in \mathbb{R}^k : \lambda \geq 0, \sum_i \lambda_i x_i = v\right\}$ is non-empty and pointed, so it has a vertex $\overline{\lambda}$. That vertex saturates k independent equality constraints among the

$3k$ constraints $\lambda \geq 0, \sum_i \lambda_i x_i \leq v, \sum_i \lambda_i x_i \geq v$. Up to reordering, we assume that $\lambda_1 = \ldots = \lambda_p = 0$ and $\sum_i \lambda_i x_{i,q} = v_q$ for a set Q of indices q with $|Q| = k - p$. The following block matrix is invertible $\begin{pmatrix} I_p & 0 \\ X_{[Q,1:p]} & X_{[Q,p+1:k]} \end{pmatrix}$, so the vectors $x_{p+1}, \ldots, x_k$ are linearly independent.

- The convex case is a consequence of the conic case.

Existence of basic LP solutions — Exercise 46

Let $Ax = b, x \geq 0$ be a system of equations and inequations. Prove that if the system is feasible, there exists a solution $\overline{x}$ such that the columns of A corresponding to positive entries of x are linearly independent.

Uses Carathéodory on $b \in \operatorname{cone}\{a_i^\top\}$.

Decomposition of polyhedra — Theorem 47

Let P be a non-empty polyhedron with $\dim(\operatorname{lin}(P)) = t$. Let one v_k be picked arbitrarily in each minimal face of P, and one r_ℓ be picked arbitrarily in each $(t+1)$-dimensional face of $\operatorname{rec}(P)$. Then $P = \operatorname{conv}\{v_k\} + \operatorname{cone}\{r_\ell\} + \operatorname{lin}(P)$. Furthermore, every Minkowski-Weyl decomposition can be written like this.

Extended formulation — Definition 48

Given a polyhedron P, a description as $P = \operatorname{proj}(Q)$ for some higher-dimensional polyhedron Q and orthogonal projection operator proj is called an *extended formulation.*

III. Perfect formulations

III.1. Integral polyhedra

Integral set — Definition 49

A convex set P is *integral* if $P = \operatorname{conv}(P \cap \mathbb{Z}^n)$.

Characterization of integral polyhedra — Theorem 50

Let P be a rational polyhedron. The following conditions are equivalent:

1. P is an integral polyhedron.
2. Every minimal face of P contains an integer point.
3. $\max_{x \in P} c^\top x$ is attained by an integral vector x whenever the maximum is finite.
4. $\max_{x \in P} c^\top x$ is integer whenever c is integer and the maximum is finite.

Proof:

1. $\Rightarrow$ 2. Let F be a face of P and $x \in F$. Since P is integral, $x = \sum_i \lambda_i x_i$ where the x_i are integral. The x_i also belong to the same face, since they must saturate the valid inequality.
2. $\Rightarrow$ 1. We can write $P = \text{conv}\{v_k\} + \text{cone}\{r_\ell\}$ while picking the vertices v_k as integral points in each minimal face. Meanwhile, the generators r_ℓ of $\text{rec}(P)$ can be chosen rational (Fourier elimination preserves it), and hence even integral too. So $P \subseteq \mathbb{Z}^n$.

III.2. Total unimodularity

Total unimodularity — Definition 51

A matrix A is *totally unimodular* if every square submatrix has determinant 0, +1 or −1.

In particular, this implies that every coefficient is in $\{0, +1, -1\}$.

Hoffman & Kruskal — Theorem 52

Let A be an integral matrix. The polyhedron $P = \{x : Ax \leq b\}$ is integral for every $b \in \mathbb{Z}^m$ if and only if A is totally unimodular.

Proof:

We only prove the "if". Let b be an integral vector and let x be a vertex of $P(b)$. Then x satisfies n linearly independent equalities among $Ax \leq b$, say $Cx = d$. We have $x = C^{-1}d$ and $C^{-1} = \left(\frac{1}{\det(C)}\right)\text{cofactor}(C)^\top$. The determinant is +1 or −1 and every coefficient in the cofactor matrix too, so $C^-1 \in \mathbb{Z}^{n\times n}$ and finally $x \in \mathbb{Z}^n$.

Integral polyhedron in general form — Exercise 53

Let A be a totally unimodular matrix. Prove that $\{x : c \leq Ax \leq d, \ell \leq x \leq u\}$ is integral for every integral (c, d, ℓ, u).

Equitable bicoloring — Definition 54

An *equitable bicoloring* of a matrix A is a partition of its columns into two sets, red and blue, such that the sum of the red columns minus the sum of the blue columns is a vector with coefficients in $\{0, +1, -1\}$.

Total unimodularity criterion — Theorem 55

A matrix A is totally unimodular if and only if it has an equitable bicoloring.

Simpler total unimodularity criterion — Proposition 56

A matrix A with coefficients in $\{0, +1, -1\}$ such that each column has at most one 1 and one −1 is totally unimodular.

Proof:

By recurrence:

- If A has a zero column, $\det(A) = 0$
- Otherwise, if A has a column with just one non-zero, we develop the determinant using that column.
- Otherwise, all columns have exactly two non-zeros so the sum of each column is 0 and $\det(A) = 0$.

III.3. Network problems

Path, flow, spanning tree, matching — Definition 57

- A *path* is a sequence of adjacent vertices.
- A *flow* is a solution to Kirchhoff's laws.
- A *spanning tree* is a tree covering every vertex.
- A *matching* is a set of pairwise disjoint edges.

Incidence matrices — Proposition 58

The edge-vertex incidence matrix of a directed graph is totally unimodular.

Graph problems with perfect formulations — Theorem 59

The following MILPs can be solved with their continuous relaxation:

- maximum flows / minimum cuts on directed graphs
- shortest paths on directed graphs
- maximum cardinality matchings on bipartite graphs

Proof:

The constraint matrix of all of these problems is a directed incidence matrix.

Using the simplex — Remark 60

The simplex algorithm must be used to obtain a basic feasible solution (a vertex). With interior point algorithms or other LP solvers, there is no guarantee that the resulting solution will be integral, hence the need for a crossover.

Exponential size of perfect formulation — Remark 61

The previous problems are exceptions: many other problems require exponentially many constraints to describe exactly the convex hull of their feasible solutions.

Spanning tree polytope — Proposition 62

The spanning tree polytope of $G = (V, E)$ is exactly described by

$$\begin{aligned} \sum_{e \in E} x_e &= |V| - 1 \\ \sum_{e \in E[S]} x_s &\le |S| - 1 \quad \forall S \subset V, S \neq \emptyset \\ x_e &\ge 0 \quad \forall e \in E \end{aligned} \tag{11}$$

Non-bipartite matching polytope — Exercise 63

Prove that the natural formulation of the maximum cardinality matching problem on general graphs is not a perfect formulation.

An equitable bicoloring of the incidence matrix corresponds to a bipartition of the nodes such that each edge has one node on either side.

Matching polytope — Proposition 64

The matching polytope of $G = (V, E)$ is exactly described by degree constraints and Blossom inequalities (see below).

IV. Valid inequalities

IV.1. Motivation

Goal of cuts — Remark 65

Given a polyhedron P, cuts are valid inequalities for the polyhedron $\operatorname{conv}(P \cap \mathbb{Z}^n)$ that are not valid for P (typically because they exclude a continuous optimal solution).

Cutting plane algorithm — Algorithm 66

Like branch and bound without branching.

Convergence of pure cutting planes — Remark 67

It doesn't always yield the convex hull of integer points in finite time!

2-variable mixed integer set — Exercise 68

Consider the set $S = \{(x, y) \in \mathbb{Z} \times \mathbb{R}_+ : x - y \le \beta\}$. Let $f = \beta - \lfloor \beta \rfloor$. Prove that $x - \frac{y}{1-f} \le \lfloor \beta \rfloor$ is a valid inequality for S. Provide a graphical interpretation.

Proof by drawing.

IV.2. Chvatal

Convex hull of integer half-space — Proposition 69

Let $H = \{x : c^\top x \leq \delta\}$ with $c \in \mathbb{Z}^n$. If the components of c are relatively prime, then $\mathrm{conv}(H \cap \mathbb{Z}^n) = \{x : c^\top x \leq \lfloor \delta \rfloor\}$.

Chvatal cut — Definition 70

A Chaval cut for the polyhedron $P = \{x : Ax \leq b\}$ is given by u where $c = A^\top u$ has integer and relatively prime coefficients, while $\delta = \lfloor b^\top u \rfloor$.

Boolean inequalities — Exercise 71

For $x \in \{0,1\}^{n+1}$, show that $\sum_{i=1}^n x_i \leq n x_{n+1}$ is equivalent to the system $x_k \leq x_{n+1}$ for all $k \in [n]$

- directly
- through Chvatal inequalities

Convergence of Chvatal cutting planes — Theorem 72

Let P be a rational polyhedron and $\mathrm{Ch}(P)$ be its Chvatal closure (the intersection of P with all Chvatal cuts). Then:

- $\mathrm{Ch}(P)$ is a rational polyhedron.
- There exists t such that $\mathrm{Ch}^{t(P)} = \mathrm{conv}(P \cap \mathbb{Z}^n)$.

This confirms the theoretical interest of cuts but doesn't tell us how to pick them.

General framework — Remark 73

Chvatal cuts are a special case of split inequalities.

Split inequality — Definition 74

An inequality $c^\top x \leq \delta$ is a split inequality if there exist $(\pi, \pi_0) \in \mathbb{Z}^n \times \mathbb{Z}$ such that $c^\top x \leq \delta$ is valid for both sets $\Pi_1 = \{x : \pi^\top x \leq \pi_0\}$ and $\Pi_2 = \{x : \pi^\top x \geq \pi_0 + 1\}$.

IV.3. Gomory

Gomory' fractional cut — Definition 75

Let $P = \{x : Ax = b, x \geq 0\}$ be a polyhedron in standard form, $u \geq 0$ be a vector of constraint weights. Denote by $c = A^\top u$ and $\delta = b^\top u$. For any $x \in P \cap \mathbb{Z}^n$, the following inequality is valid:

$$\sum_i (c_i - \lfloor c_i \rfloor) x_i \geq \delta - \lfloor \delta \rfloor \tag{12}$$

Proof:

For an integer solution x, the term $\sum_i \lfloor c_i \rfloor x_i$ is integer, and so is $\lfloor \delta \rfloor$. Therefore, subtracting them on both sides leads to

$$\sum_i (c_i - \lfloor c_i \rfloor) x_i = \delta - \lfloor \delta \rfloor \mod 1 \tag{13}$$

Both terms are positive and the right-hand side is in $[0, 1)$, hence the inequality.

Gomory weights — Remark 76

The weights can be chosen to exclude a basic optimal solution $\overline{x}$ to the continuous relaxation.

Let $\overline{x}$ be associated to simplex basis B: we have $x_B = A_B^{-1}(b - A_N x)$. The i-th row of that system can be written $x_i + \sum_{i \in N} c_i x_i = \delta$ where $c = e_i A_B^{-1}$. The corresponding Gomory cut is $\sum_{i \in N} (c_i - \lfloor c_i \rfloor) x_i \geq \delta - \lfloor \delta \rfloor$, where we only sum over non-basic indices.

Taking i such that $\overline{x}_i \notin \mathbb{Z}$ and remembering that $\overline{x}_i = \delta$, we see that the Gomory cut excludes $\overline{x}$ since $\overline{x}_N = 0$ and $\delta - \lfloor \delta \rfloor > 0$.

IV.4. Structured approaches

Blossom inequality for matchings — Exercise 77

Prove that the characteristic vector of a matching on $G = (V, E)$ satisfies $\sum_{e \in E[U]} x_e \leq \frac{|U|-1}{2}$ for every subset $U \subseteq V$ of odd cardinality.

Let M be a matching and $e \in M \cap E[U]$: e covers two nodes of U. However, each node of U is covered by at most one edge of M, so $|\ M \cap E[U]\ | \leq |U\frac{|}{2}$. When $|U|$ is odd, we can round down.

Cover inequality for knapsack — Exercise 78

A subset of indices C is a *cover* for the knapsack set $K = \{x \in \{0,1\}^n : a^\top x \leq b\}$ if it satisfies $\sum_{i \in C} a_i > b$, and it is *minimal* if $\sum_{i \in C \setminus \{j\}} a_i < b$ for all $j \in C$. The inequality $\sum_{i \in C} x_i \leq |C| - 1$ is valid for any cover. How can we further strengthen it for minimal covers?

The knapsack cannot contain every element of a cover. We can even strengthen the cut as follows: let $E = \{k : a_k \geq \max_{i \in C} a_i\}$, then

$$\sum_{i \in C} x_i + \sum_{k \in E} x_k \leq |C| - 1 \tag{14}$$

Useful beyond knapsack — Remark 79

These cuts apply to any single constraint of a binary integer program.

Comb inequality for TSP — Exercise 80

Billera-Sarangarajan — Theorem 81

Any $\{0,1\}$ polytope is affinely equivalent to a face of an asymmetric TSP polytope in sufficiently large dimension.

Bibliography

[1] M. Conforti, G. Cornuéjols, and G. Zambelli, *Integer Programming*, 1st ed. 2014., no. 271. in Graduate Texts in Mathematics. Cham: Springer International Publishing : Imprint: Springer, 2014. doi: 10.1007/978-3-319-11008-0.

[2] S. Gaubert and F. Bonnans, *Recherche opérationnelle: Aspects mathématiques et applications*. Editions de l'école polytechnique, 2016.